

PATARI v0.3 Instructions

Analysis Tool for Clinical Photoacoustic Imaging

<https://github.com/Mo-Sc/PATARI>

UNTESTED VERSION – USE ONLY FOR EVALUATION PURPOSES

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- Description
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- An open source GUI tool that covers the most common analysis workflows for clinical photoacoustic studies
 - Target group: medical doctors / non-technical users
 - Pure-python based
 - Runs on Windows / MacOS / Linux
 - No additional dependencies, easy installation
 - Relies on established frameworks:
 - NAPARI (GUI)
 - PATATO (Data I/O, Processing)
 - Modular architecture:
 - Can be adapted to individual needs
 - As simple as possible, as complex as necessary

- Supports different file formats + easy extension to new ones
 - Native iThera (including iLabs annotations)
 - PATATO HDF5
 - IPASC, SIMPA (not tested)
- Browse studies and fast switching between scans
- Overlay / side by side view of US and PA images
- Seamless scrolling through frames and wavelengths
- Spectral unmixing, including chromophore spectra and THb / sO2 calculation (provided by PATATO)
- Postprocessing / reconstruction (provided by PATATO) (not included in v0.3)
- Export processed scans to HDF5
- **Quantitative image analysis:**
 - Draw, edit, and save ROIs of arbitrary shapes
 - Import / export of ROIs , create a custom ROI Library
 - Extract custom statistical features from ROIs, auto-updating analysis table and XLSX export
 - Fast intensity extraction over multiple layers, frames, wavelengths or chromophores.
 - Visualize ROI intensities over time and spectrum, plot histograms.
- **Automation:**
 - Deep learning based tissue segmentation from US image (<https://doi.org/10.1117/12.3049067>)
 - Automatic ROI placement in target class (i.e. muscle, fat layer, ...)
 - Batch processing mode (not included in v0.3)

1. Download the latest release for your operating system (macOS or windows)
2. unpack the zip folder
3. Run the patari file:
 1. On first run: If you get an error, stating PATARI could not be verified, give your OS permission to run it:
 - For windows: Click `More Info` -> `Run Anyways`
 - For macOS: Open `System Settings` -> `Privacy & Security` -> Scroll down to Security -> “patari was blocked” -> `Open Anyway`
 2. PATARI will now download all the required files, so make sure your computer is connected to the internet. The installation can take a few minutes.
4. Once the installation is done, PATARI launches automatically. If patari is closed, you can simply launch it again by clicking on the same patari file.

- At installation, PATARI creates a *.patari* folder in the users home directory and copies all the relevant configuration files into it. It also contains a *models* folder, where PATARI auto-downloads pretrained segmentation models, once the segmentation adapter is run for the first time.
- To change settings in PATARI, edit the respective .json file (requires restart):
 - *config.json*: main configuration file. Contains general settings, like which features are displayed in the tables, as well as task-specific settings for the different modules.
 - *roi_library.json*: Contains all the ROIs in the ROI Library. Entries can be manually added or deleted.
 - *segmentation_models.json*: Configuration for the automatic segmentation. See segmentation reference for more info (tbd).
- In a future version, these settings will be editable from the GUI as well.
- **Keyboard shortcuts:**
 - It is recommended to use keyboard shortcuts for repetitive commands. Details can be found in the *PATARI_Shortcuts.pdf* document.

Usage

Image Viewer & Navigation

Disclaimer: Some of the screenshots might be from an earlier version and therefore look slightly different.

Usage

Overview (widget positions can differ)

Layer Settings Dock

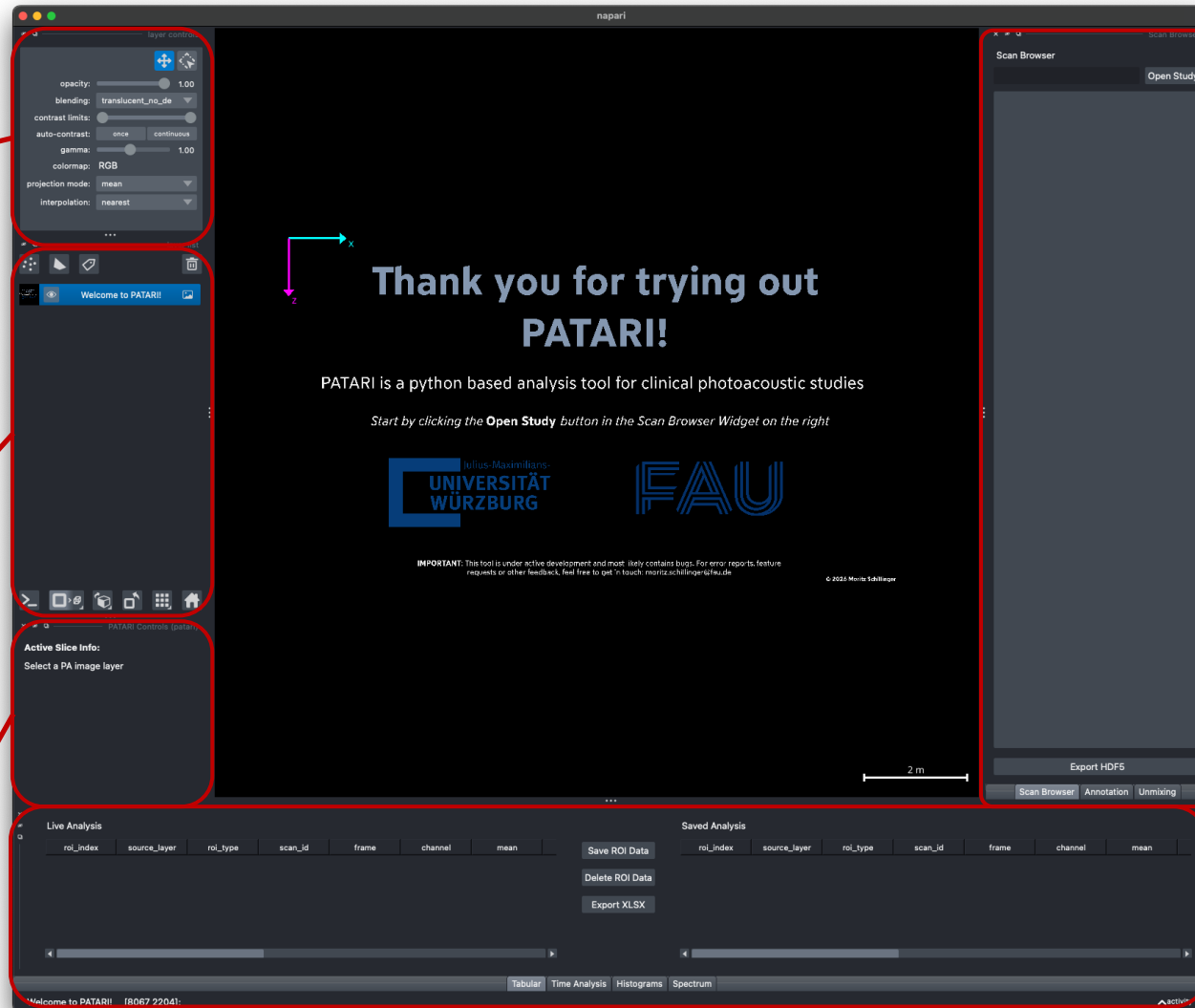
Image layers: control contrast, colormap, blending/opacity: *multiplicative* for automatic blending
translucent_no_depth for manual control via opacity slider
ROI Layer: Manually define ROIs to place in scan

Layer List

List of all layers included in the scan + ROI layer
Selecting a PA makes it the **Active Layer**. All other PA layers are automatically set to hidden (eye button)
Processing and ROI analysis is always performed on the **Active Layer**

Slice-Info Dock

Shows relevant information about the selected slice
A slice refers to a single 2D image of a given layer, frame and channel (wavelength or chromophore)

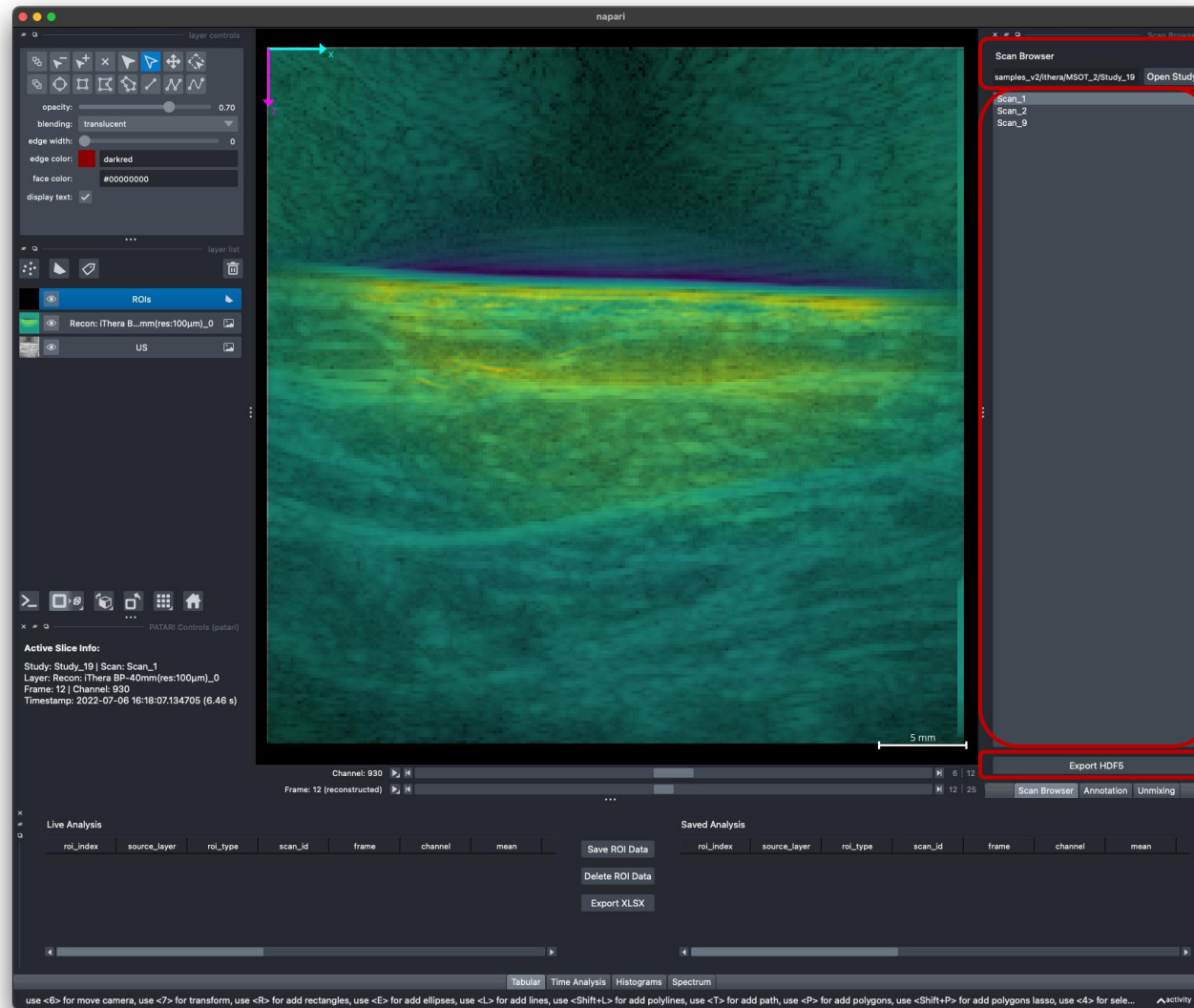


Control Dock

Scan Browser: Open Studies, switch Scans, export scans to HDF5
Annotation: ROI-Library, ROI settings
Unmixing: Create chromophore layers

ROI-Analysis Dock

Extracts the data within the ROIs from the **Active Layer**
Tabular: Calculates and collects statistical features in the ROIs
Time Analysis: Displays mean intensity in the ROIs over time
Histogram: Shows histogram of ROI intensities at selected frame and channel
Spectrum: Shows ROI intensities over channel (wavelength or chromophore)



Open study folder or individual scan folder

If multiple scans are present in the study folder, they will be listed here

By default, first scan will be opened

Switch to other scan by clicking on the scan name

For large scans, opening might take a few seconds

Exports the opened scan into a HDF5 file (see [here](#) for more info)

Includes all data generated by the plugin (currently ROIs and unmixed datasets)

Exported HDF5 files can be opened using this plugin again

Overwrite policy:

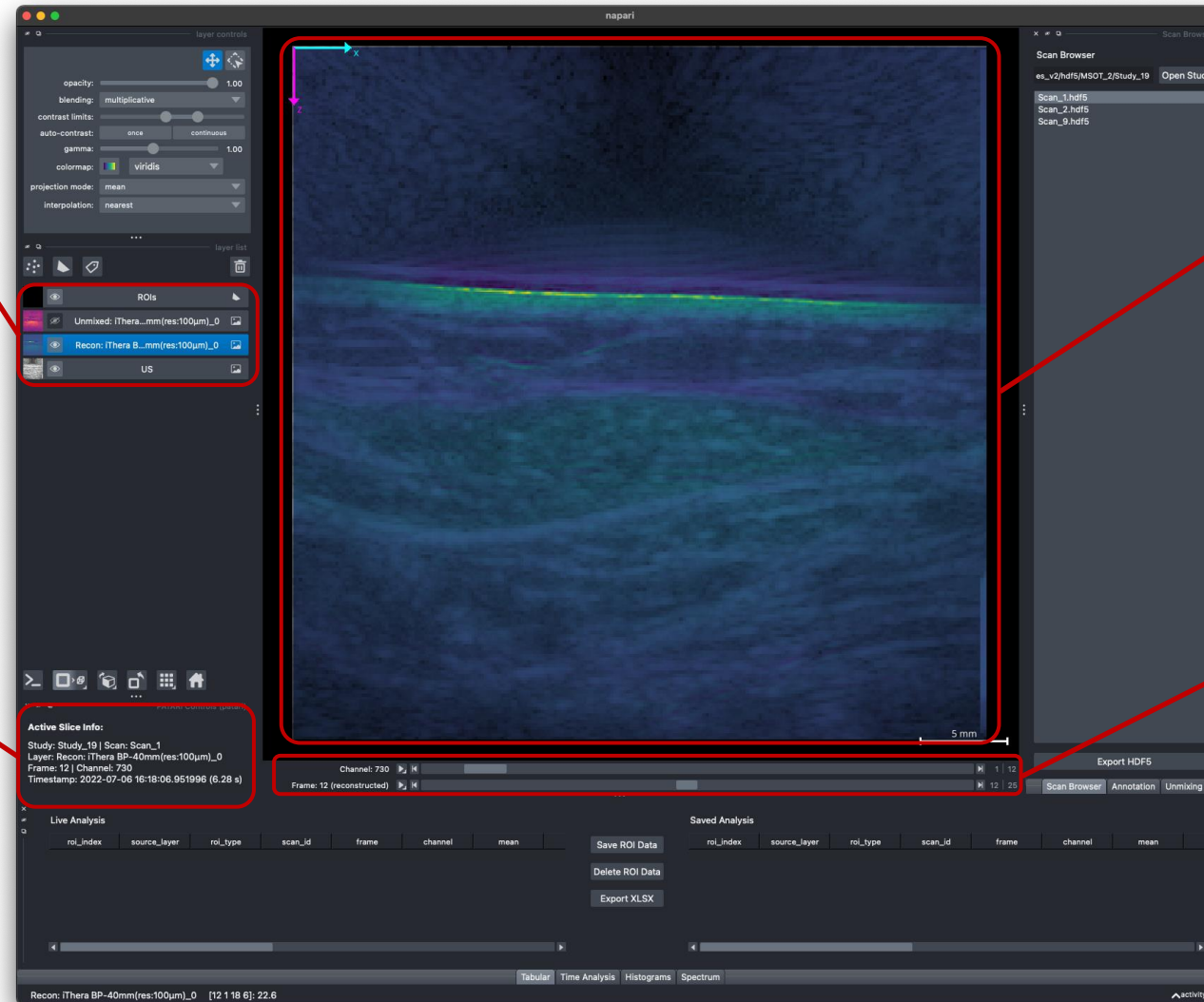
- Data always has to be exported to a new file (cant append to existing HDF5 file)
- Data that was in the original file **always persists**
- Data that was previously generated by PATARI is **always overwritten**

Usage

Image Viewer

This layer list contains two PA layers (Recon & Unmixed). The Recon layer is selected as active layer and will therefore be used for processing and analysis.

The Slice-Info Dock also shows the active layer, selected frame and channel, as well as the timestamp (if available).



Main Image Viewer

By default, the viewer shows the ultrasound layer, overlaid with a reconstructed PA layer, and the ROI layer.

This can be changed in the layer list and the layer settings.

The two sliders below the viewer are used to select the frame and the channel. Channel can be either wavelengths or chromophores.

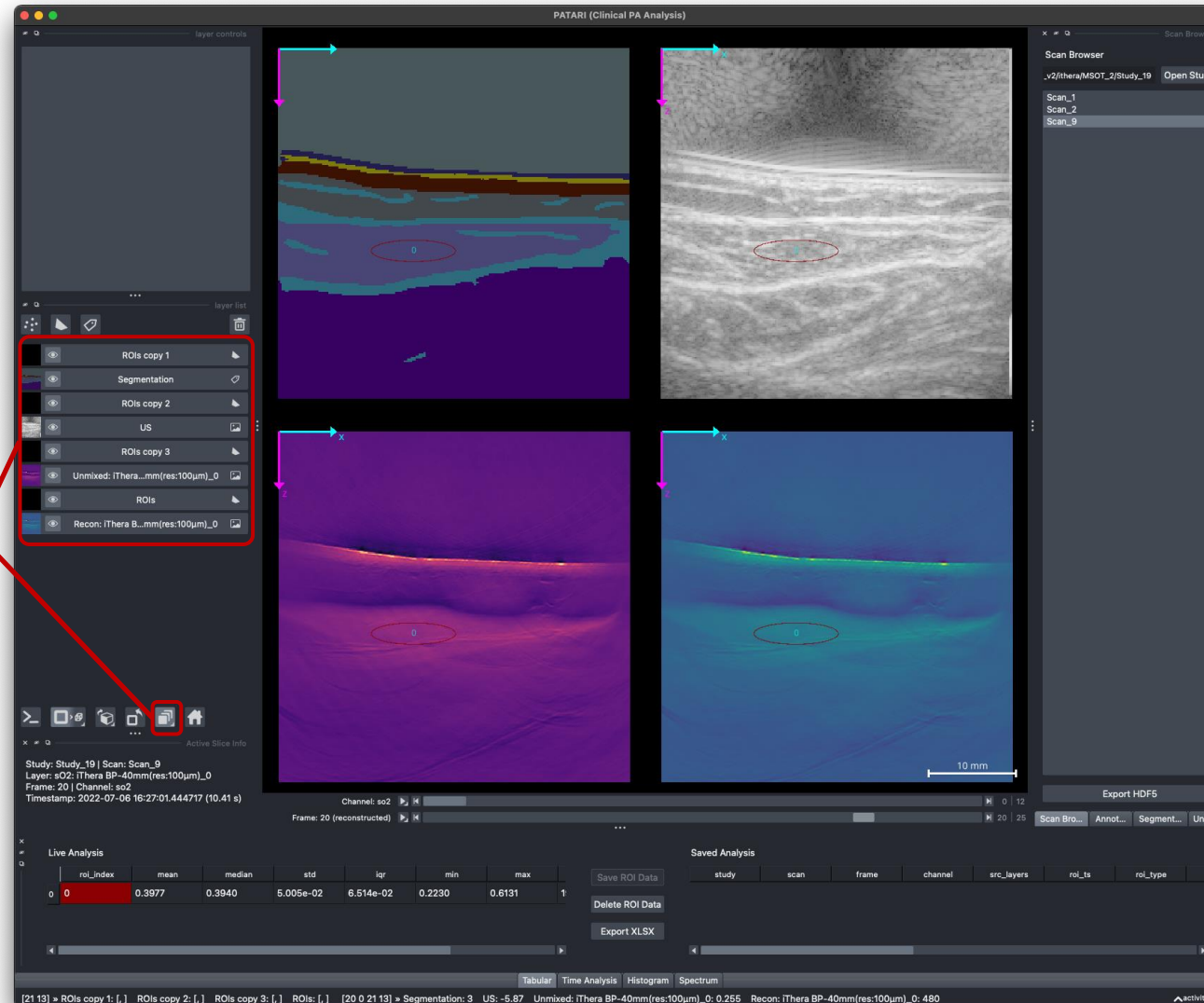
The indices are synced across all layers. The available indices and the slider labels depend on the active layer.

Image Viewer – Grid Mode

Grid Mode

Clicking on the grid mode button displays the layers next to each other instead of an overlay.

By default, always two layers in the layer list are overlaid with each other



Limitations

The Grid Mode is not a true multi-viewer feature. It can be used to view different layers next to each other. But it is not possible to change the axes (frame / channel) independently for each grid view. It is also not suitable for annotation, since the ROI layer is not synced across the grid.

A workaround to at least show the ROI layer on all shapes is to duplicate the ROI layer (right click → duplicate) and arrange the layers so that before every image layer there is a ROI layer (because every 2 consecutive layers are always overlaid), as shown in the screenshot. But keep in mind that the Analysis table is only connected to the one original table.

This is a [limitation of the underlying napari framework](#), but will hopefully be solved in future versions.

Alternative

It is also possible to launch PATARI twice to have completely disentangled windows, for example to compare different scans. But keep in mind that this means ROIs, data or settings are not synced between the instances.

Usage

Annotation & Analysis

Disclaimer: Some of the screenshots might be from an earlier version and therefore look slightly different.

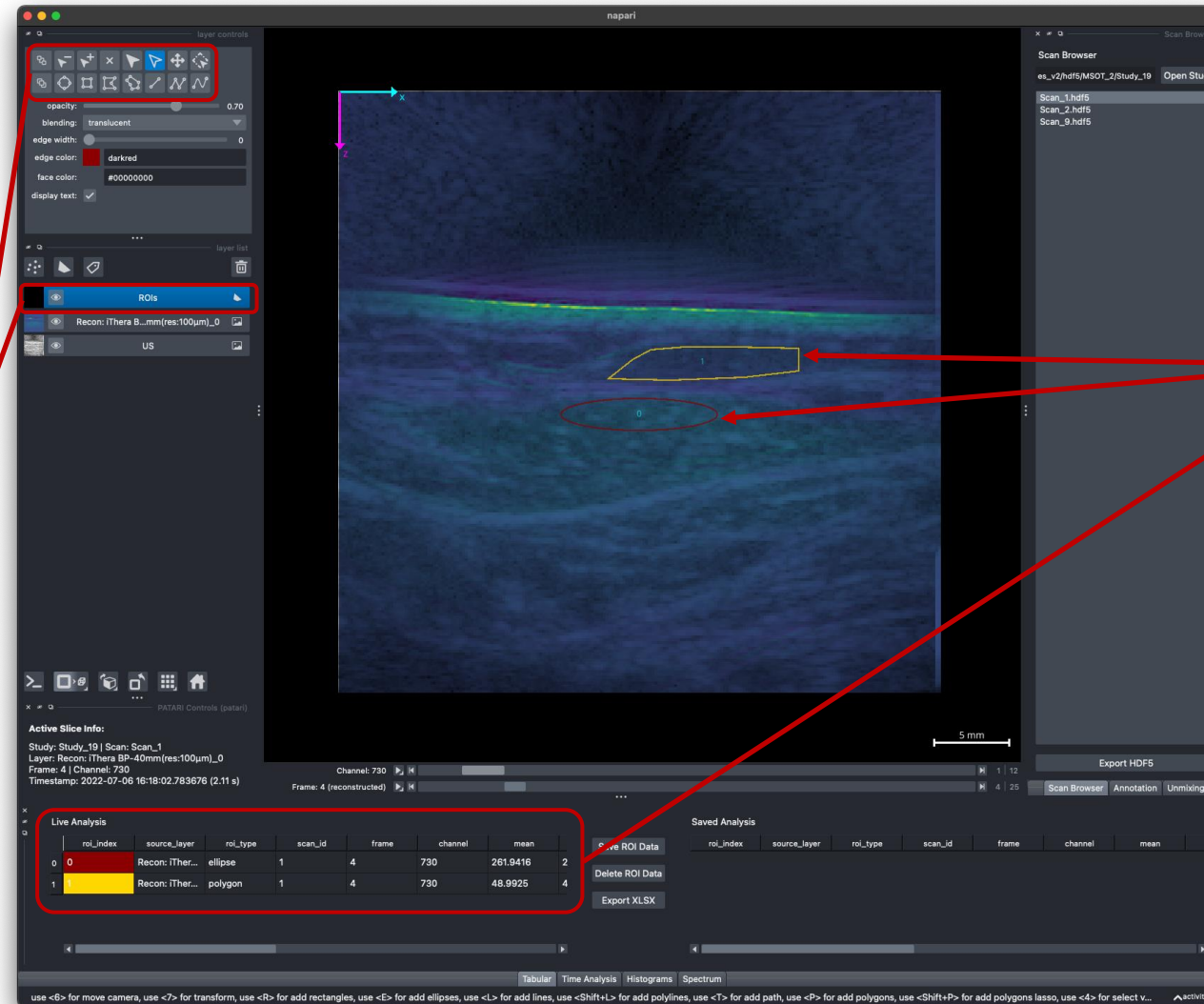
Manual ROI Drawing

To start drawing ROIs, select the **ROIs layer** in the layer list.

This will show the ROI layer controls in the layer settings:

- Select  or press **E** to draw an ellipse
- Select  or press **R** to draw a rectangle
- Select  or press **P** to draw an ellipse
- Select  or press **S** to select, move, or resize an existing ROI
- Select  or press **D** to edit polygon ROIs
- Select  or press **DEL** to delete the selected ROI

Note: They keyboard shortcuts only work when the ROIs layer is selected



New ROIs are automatically added to the **Live Analysis Table**.

To remove a ROI from the table, delete it in the viewer.

ROIs can be identified by **ROI ID** and **colour**. The ROI features in the Live Analysis Table are **automatically updated** when switching between active layers or scrolling through wavelengths / channels.

To select a ROI, select the ROIs Layer and use the select tool (shortcut **S**) and click on the ROI. Newly drawn ROIs are automatically selected.

Live Analysis Table

This table lists all the ROIs that are currently drawn in the viewer.

Imported ROIs (i.e. created by iLabs and included in the file) are also displayed here.

The extracted ROI data auto-updates when layer, frame or channel change.

ROIs are scan-specific, and are dropped when a new scan is opened.

The ROIs in the Live Table are written to file, when the scan is exported using the Export HDF5 button in the scan browser.

Save ROI Data: Copies the selected ROI (here ID 1) from the Live Table to the Saved Table

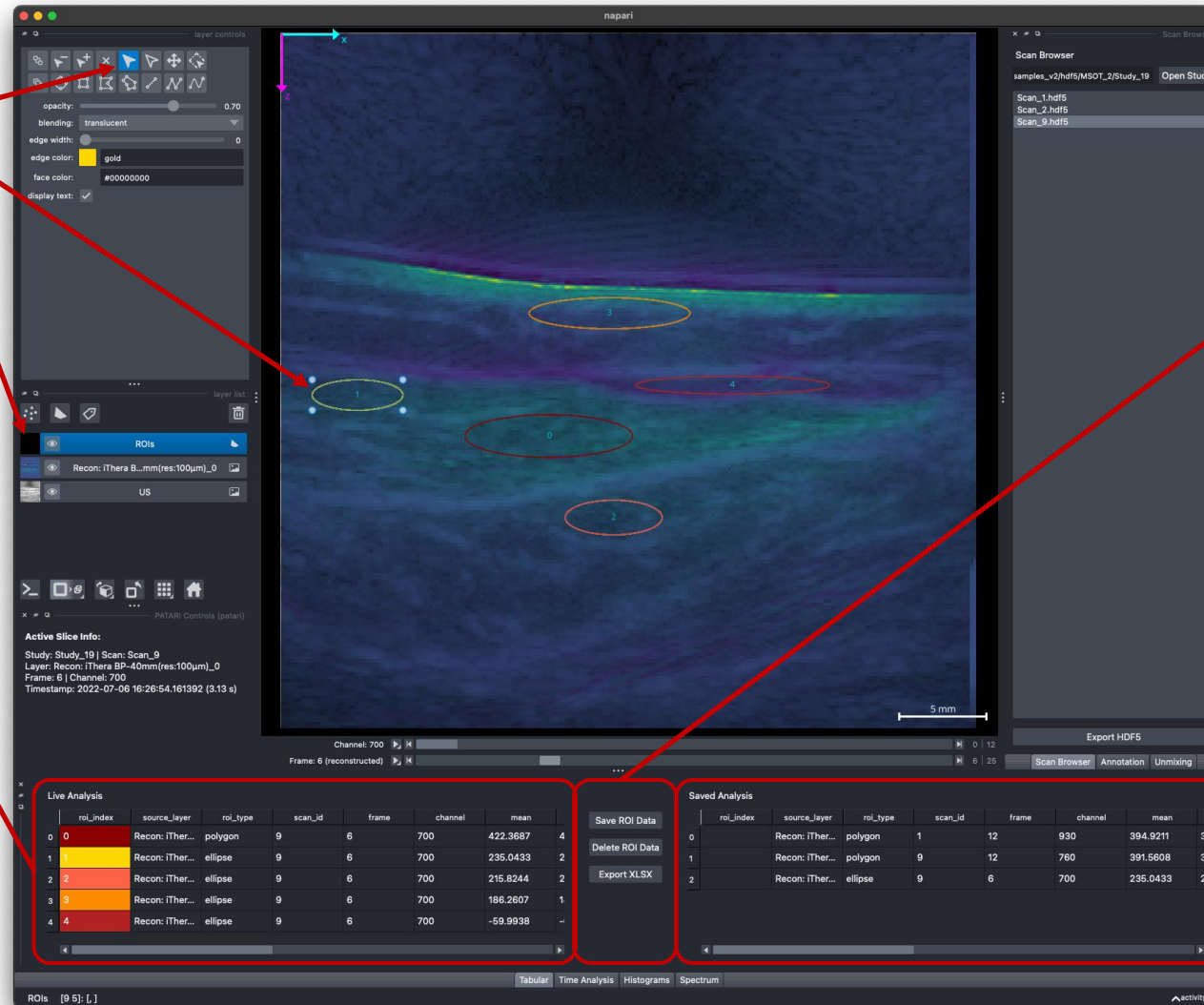
Delete ROI Data: Removes the selected row(s) from the Saved Table

Export XLSX: Exports the Saved Table into a MS Excel file.

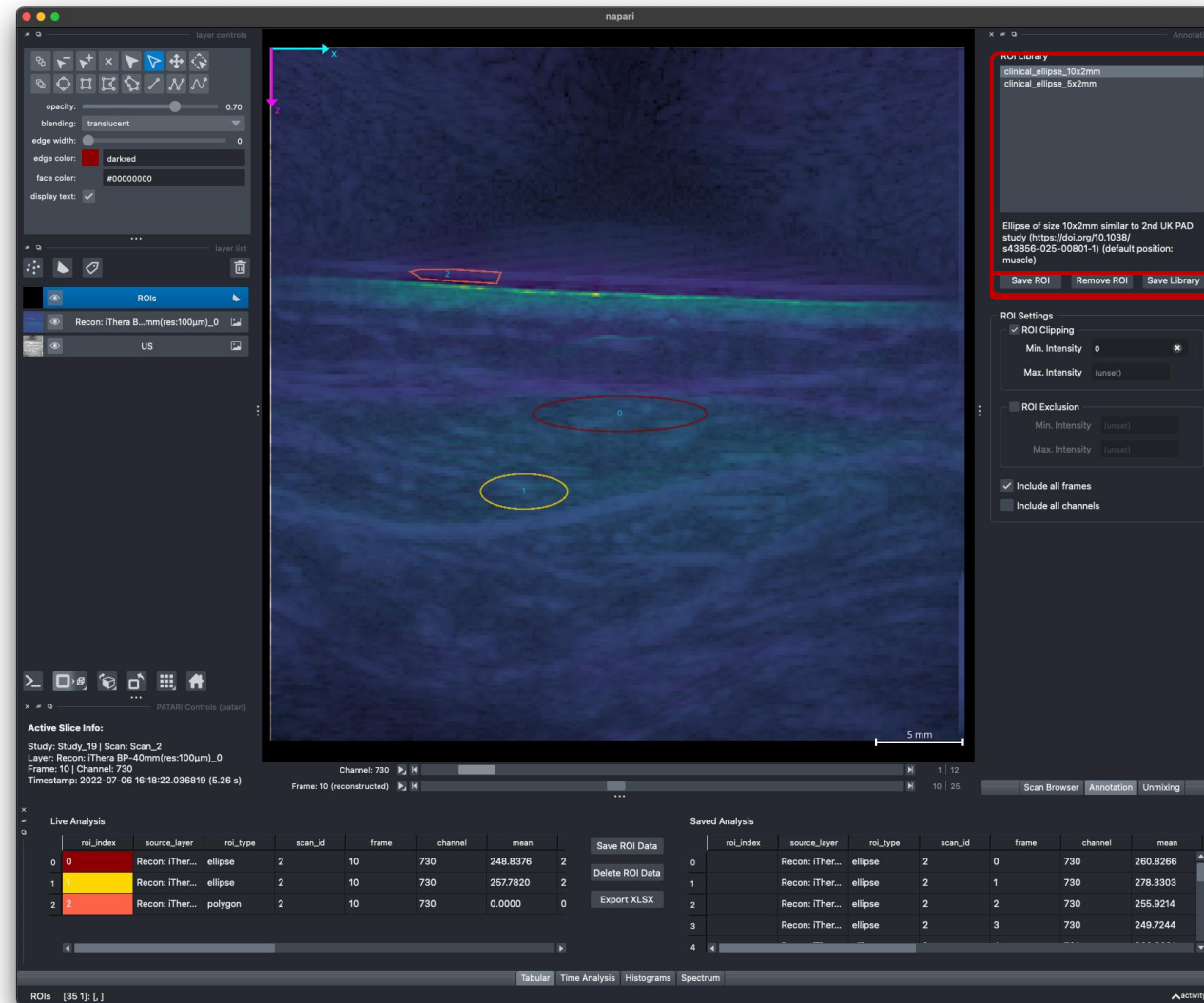
Saved Analysis Table

Once a Live Table entry was added to the Saved Analysis Table, it does not change anymore and is saved, even when a new scan or study is opened.

Example: 3 ROIs were saved. One ROI (3rd row) was added from the current Live Analysis Table. But the other two were saved from a different wavelength (2nd row) or even a different scan (1st row).



Annotation Widget – ROI Library



ROI Library

Contains preconfigured ROI shapes.

Can be used to quickly draw identical ROIs across multiple studies and scans.

Clicking on a ROI will show its description.

Double-clicking will place the ROI into the viewer (and add it to the Live Table).

ROIs can be added or removed to/from this library.

Added entries will persist as long as the app runs.

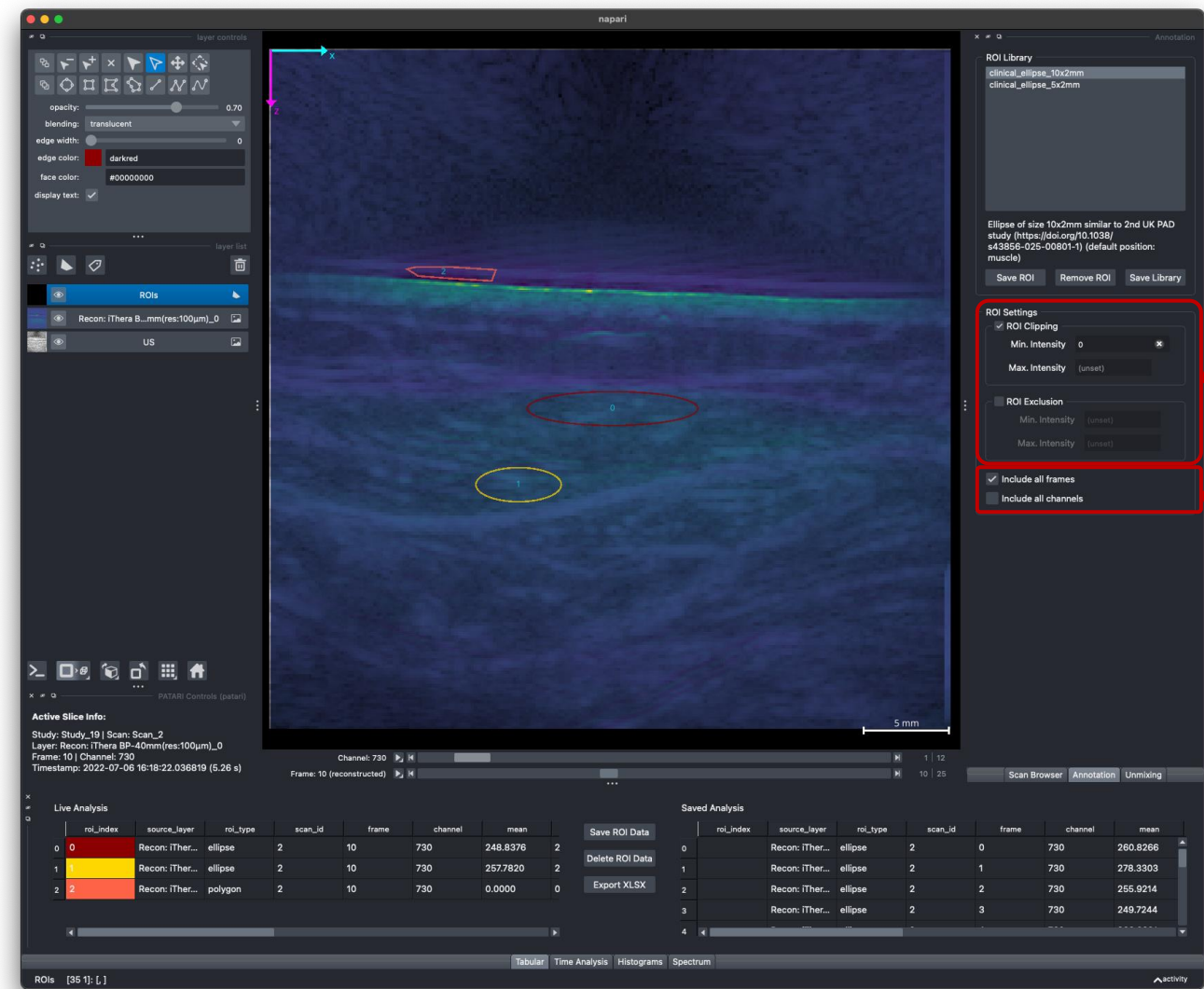
Library is connected to a ROI file, so ROIs can be shared across devices.

Save ROI: Adds the ROI shape that is selected in the viewer to the library.

Remove ROI: Removes the ROI that is selected in the library from the library.

Save Library: Permanently saves the ROI Library to file so it persists even when the app is closed.

Annotation Widget – ROI Settings



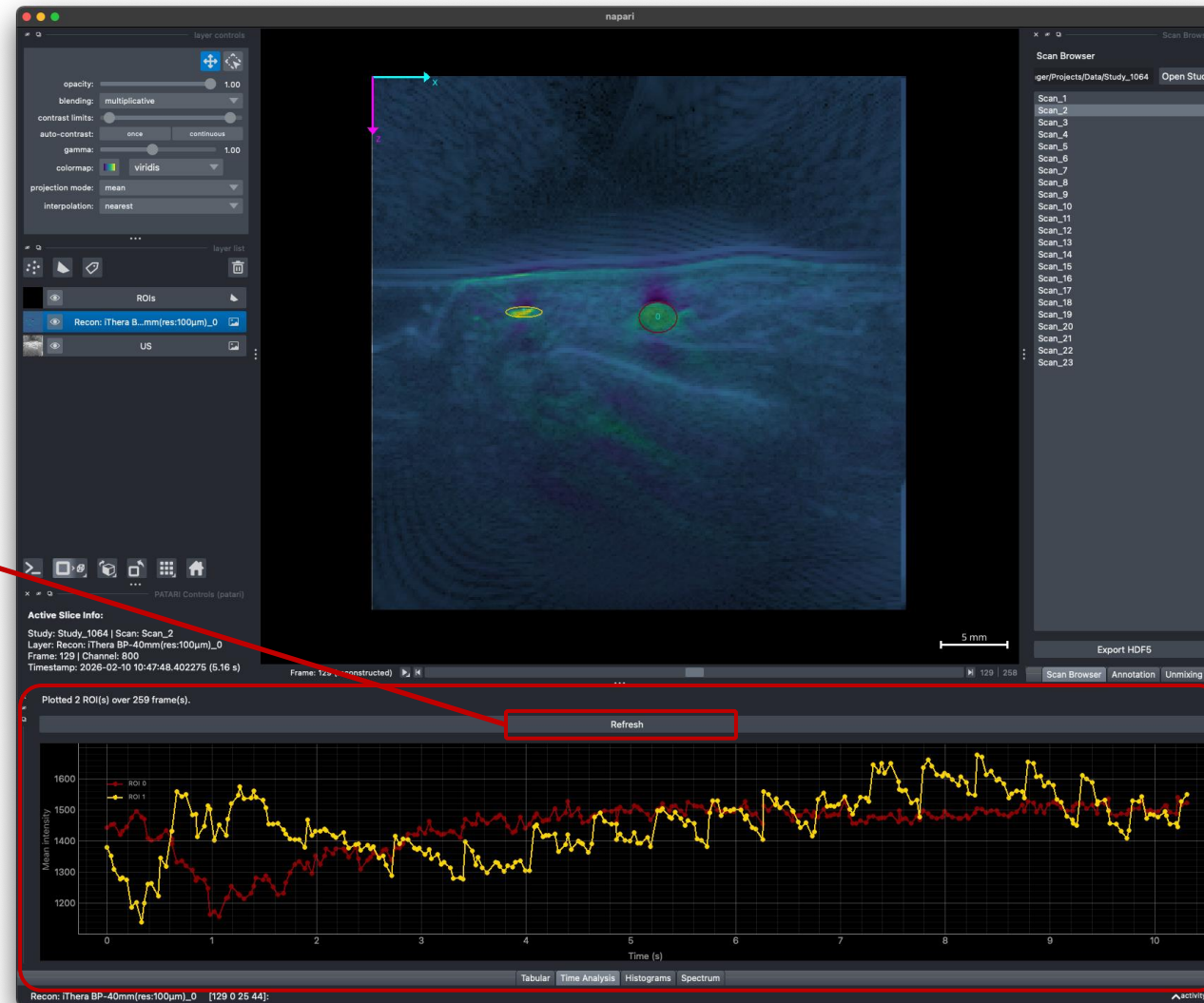
Settings to control which intensities are included in the ROI feature calculations. Settings are automatically applied to all the ROIs in the Live Table

ROI Clipping: Clips all the intensities in the ROIs to the given range

ROI Exclusion: Excludes all the intensities outside the given range from the feature calculation

Controls the behaviour of the **Save ROI Data** button

If checked, saving an ROI will add entries for all the frames and/or channels of the active layer to the Saved Table, instead of just the selected slice



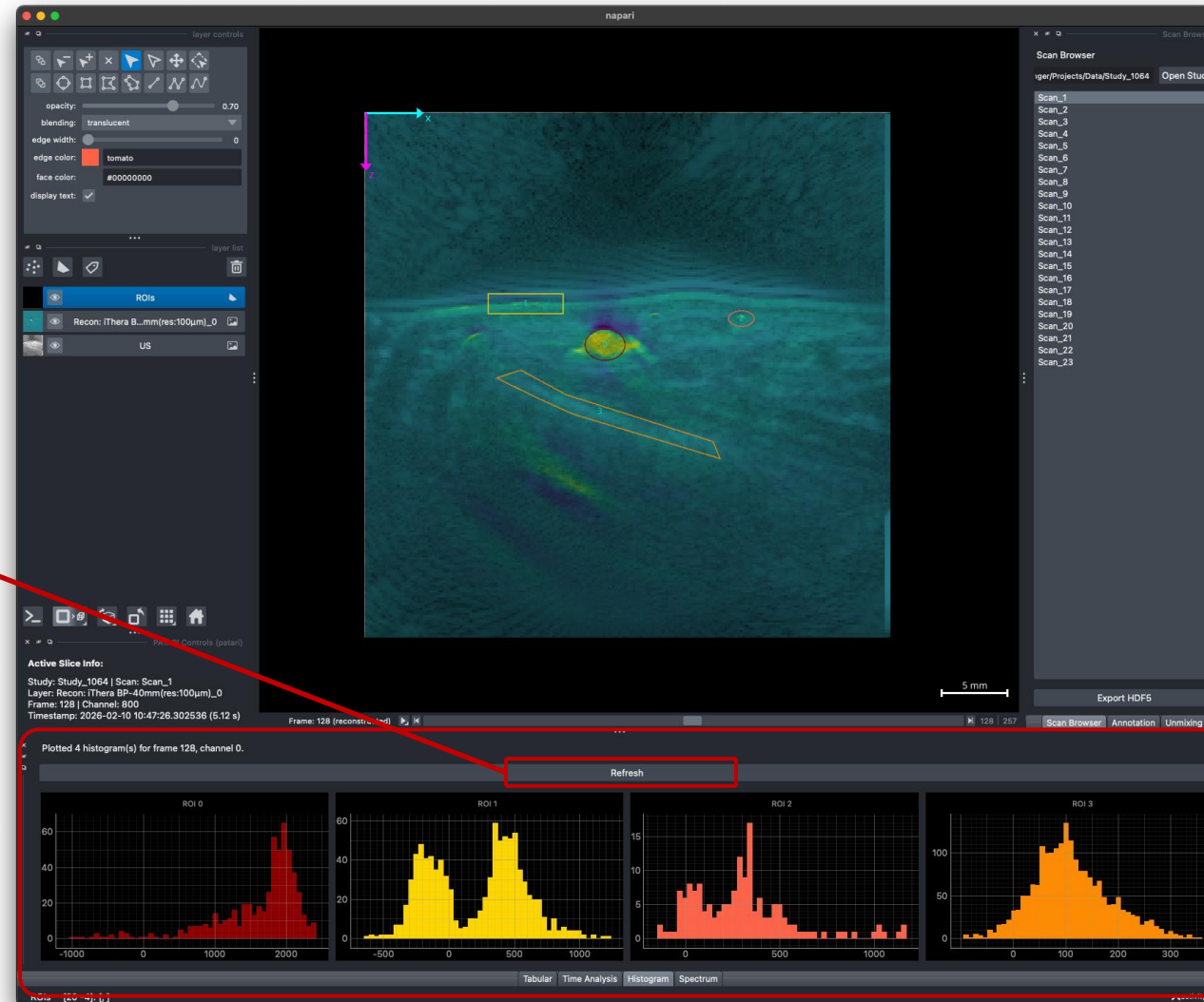
Unlike the Live Table, plots are not auto-updated. When ROIs, layer or channel are changed, **Refresh** has to be clicked again.

Plots ROI intensities over time.

y-axis: mean intensity in the ROI for the selected channel.

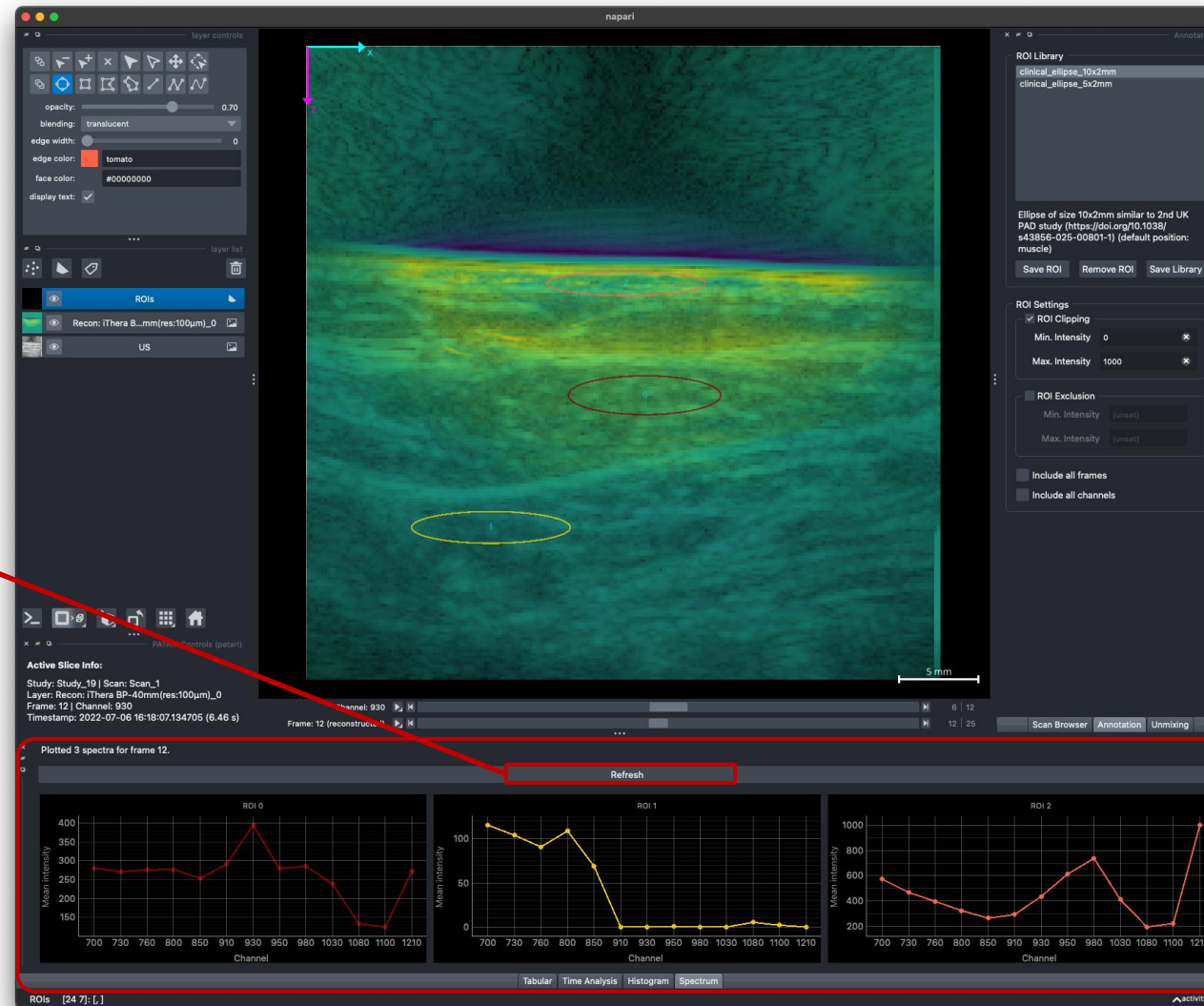
x-axis: if timestamps are available, relative time since the first frame. Otherwise frames.

Histogram Analysis



Unlike the Live Table, plots are not auto-updated. When ROIs, layer, frame or channel are changed, **Refresh** has to be clicked again.

Plots a histogram for each ROI, showing the distribution of intensities



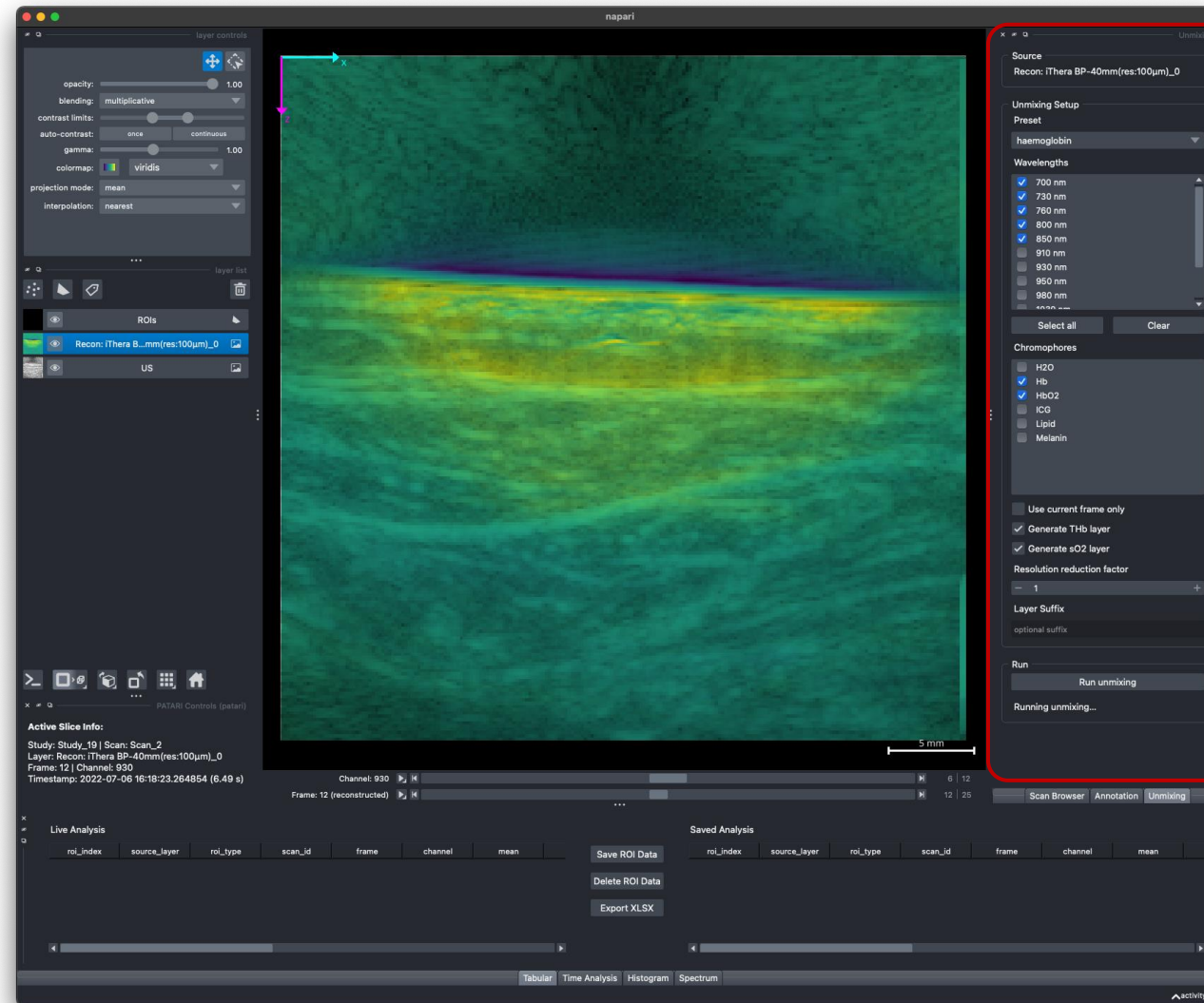
Unlike the Live Table, plots are not auto-updated. When ROIs, layer or frame are changed, **Refresh** has to be clicked again.

Plots a spectrum for each ROI, showing the intensities in the ROI across the different channels (wavelengths or chromophores)

Usage

Image Processing

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Unmixing Widget

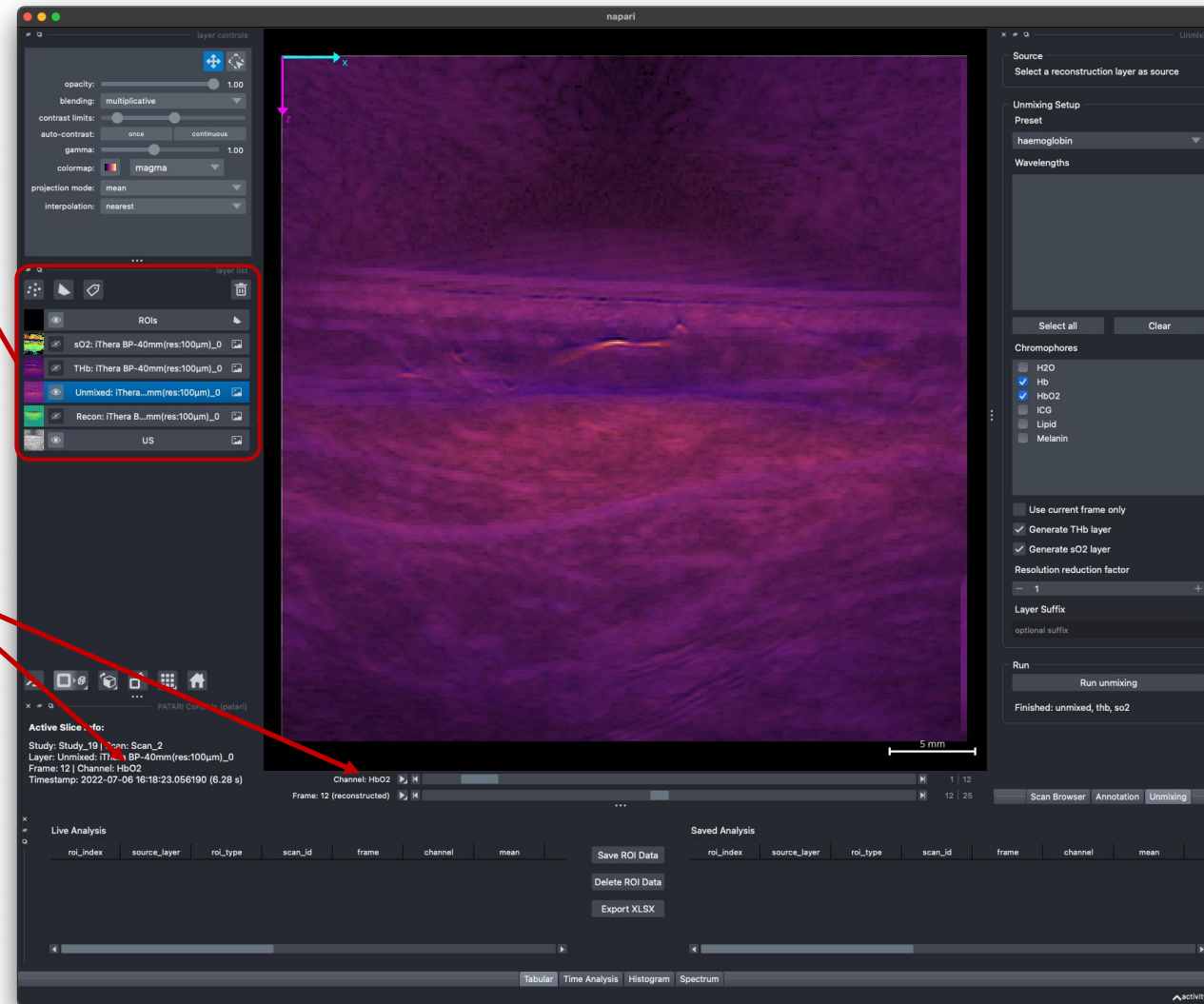
1. Unmixing is performed on the active layer. The active layer must be a PA reconstruction with multiple wavelengths.
2. Select wavelengths to use and target chromophores.
Alternatively, use one of the given presets. Presets can be defined and shared using json files.
3. Select whether all frames of the active layer or just the currently selected frame should be unmixed.
4. Select whether total hemoglobin (THb) and oxygenation (sO2) should also be calculated (requires Hb and HbO2 selection).
5. Choose a resolution reduction factor (1 = no reduction).
6. Add a custom suffix that will be added to the unmixed layers name.
7. Click Run unmixing (can take a few seconds, depending on number of wavelengths and image dimensions).

Unmixing – Layers

Unmixed Layers

Unmixed scans are added as additional layers to the layer list, and can be selected as active layers. If selected, THb and sO2 layers are added as well.

Unmixed layers have chromophores as channel dimension

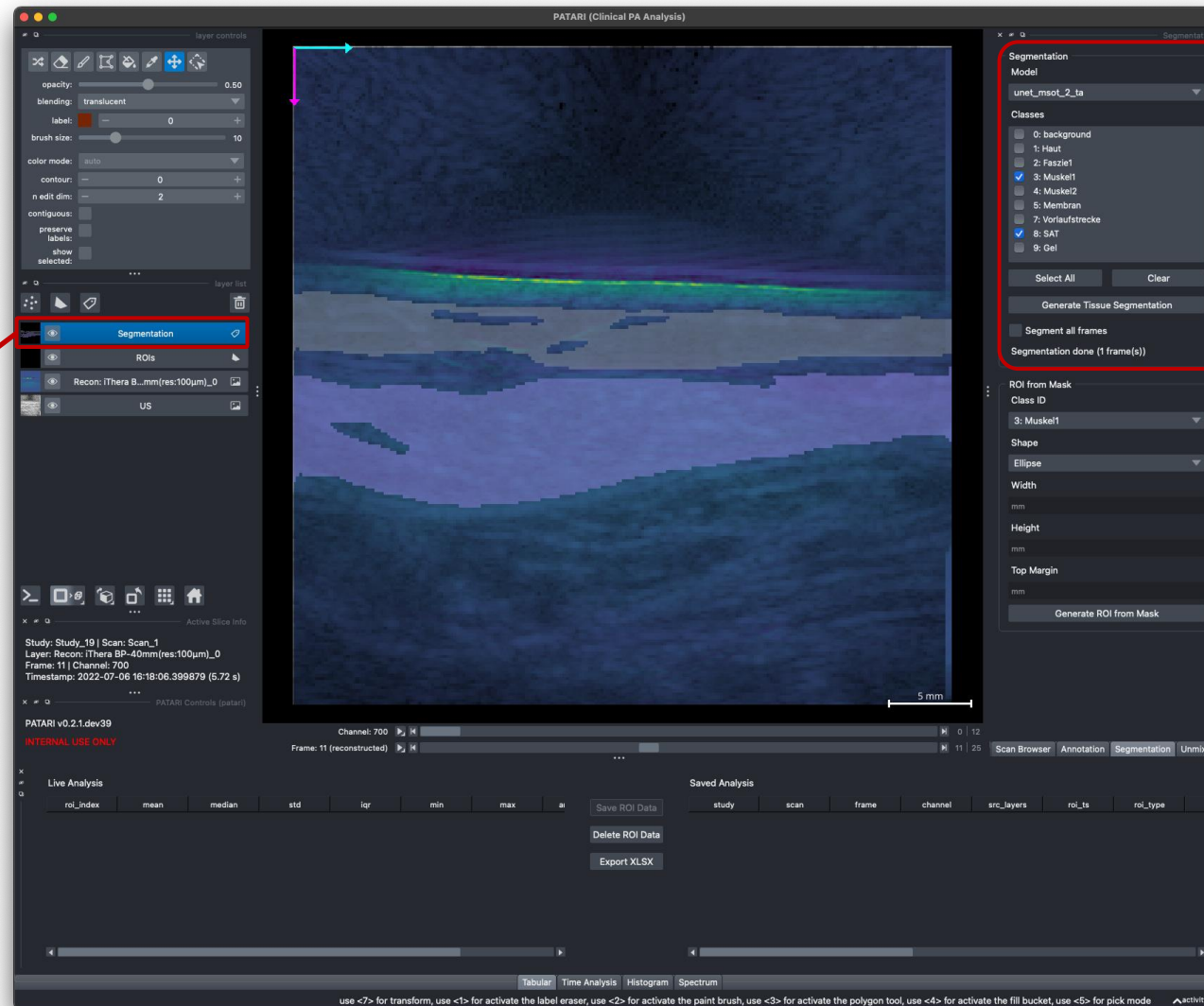


Usage

Segmentation & Auto-ROI

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The segmentation is added as an additional layer in the layer list



Segmentation

Model: Select the segmentation model for your task.

Classes: The segmentation classes that are provided by the selected model. You can select which classes should be included in your tissue segmentation map.

Generate Tissue Segmentation: Create the segmentation map. It will appear on the layer list on the left.

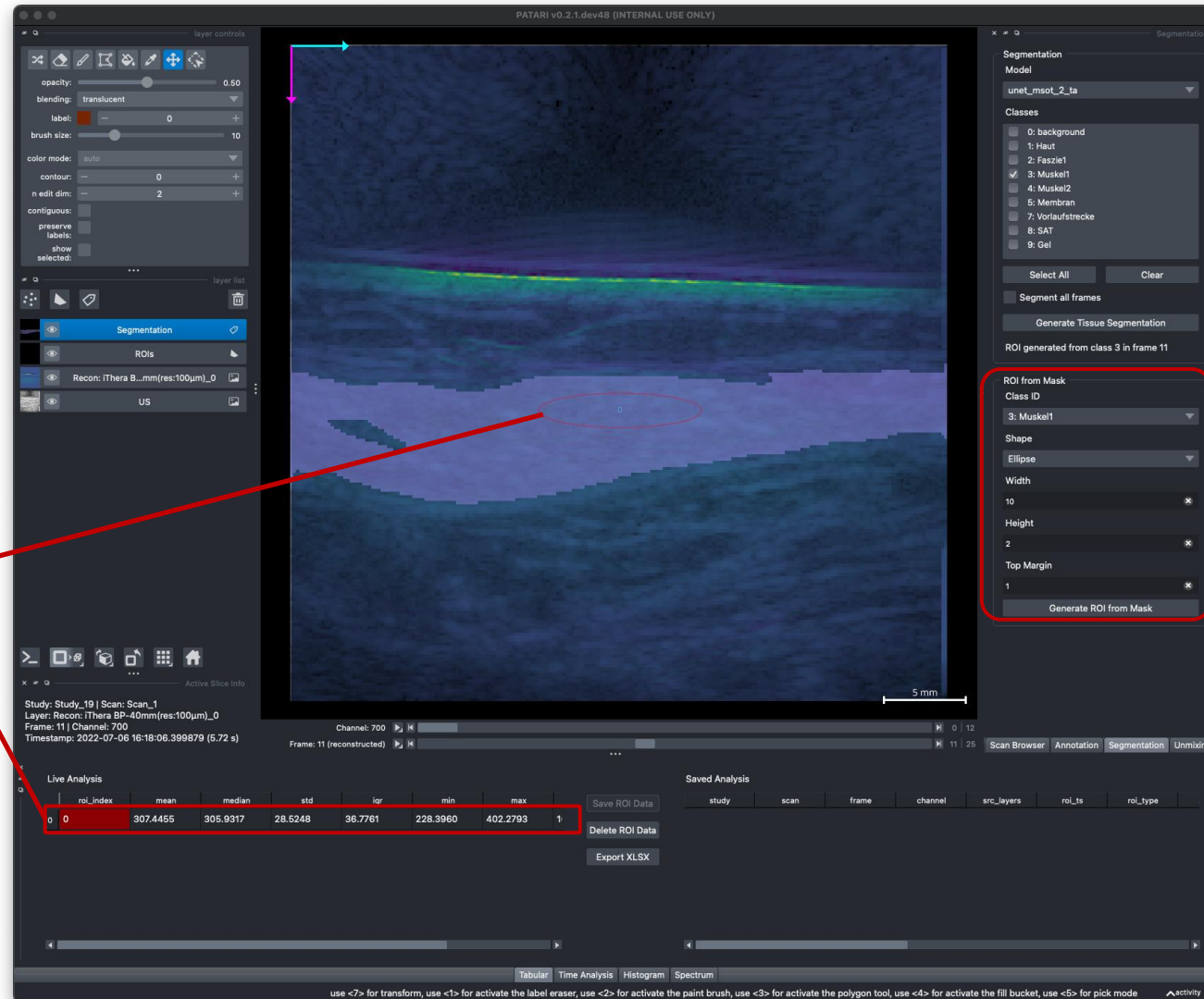
Segment all frames: By default, the model only creates a mask for the currently selected frame. If this box is checked, it will segment all the frames. This can require more time, depending on the number of frames.

Note: If a model is used for the first time, its weights are downloaded automatically from the internet. This can take some time, depending on the connection

Automatic ROI Placement

From segmentation widget (Ellipse and Rectangle shapes)

The newly generated ROI appears automatically in the viewer and the Live Analysis Table. It can be moved and edited freely, like a manually created ROI



Auto ROI

If a segmentation mask exists, it can be used to automatically generate a ROI in the target muscle

Class ID: The tissue class where the ROI should be placed in.

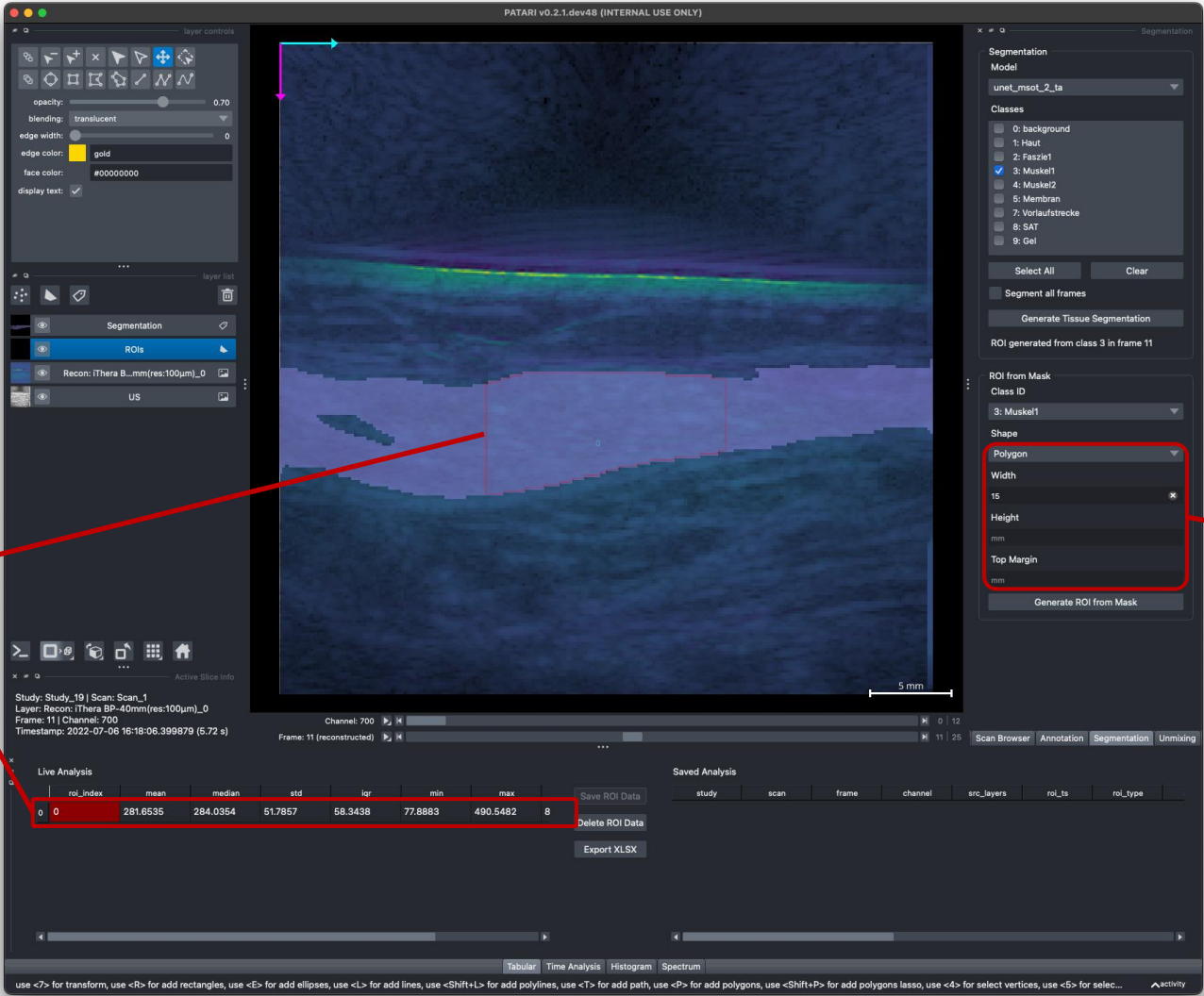
Shape: The shape of the generated ROI. Can be Ellipse, Rectangle or Polygon (see next slide).

Width + Height: Size of the generated ROI.

Top Margin: A margin between the upper end of the target class and the beginning of the generated ROI

Automatic ROI Placement

From segmentation widget (predefined shapes)



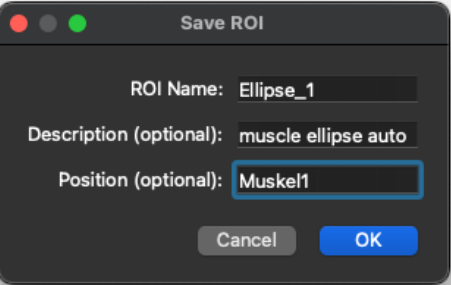
The newly generated ROI appears automatically in the viewer and the Live Analysis Table. It can be moved and edited freely, like a manually created ROI

Auto ROI
If Polygon shape is selected, the ROI will follow the outline of the target class in the segmentation mask. Height and width can be used to center-crop the outline to the given size.

Automatic ROI Placement

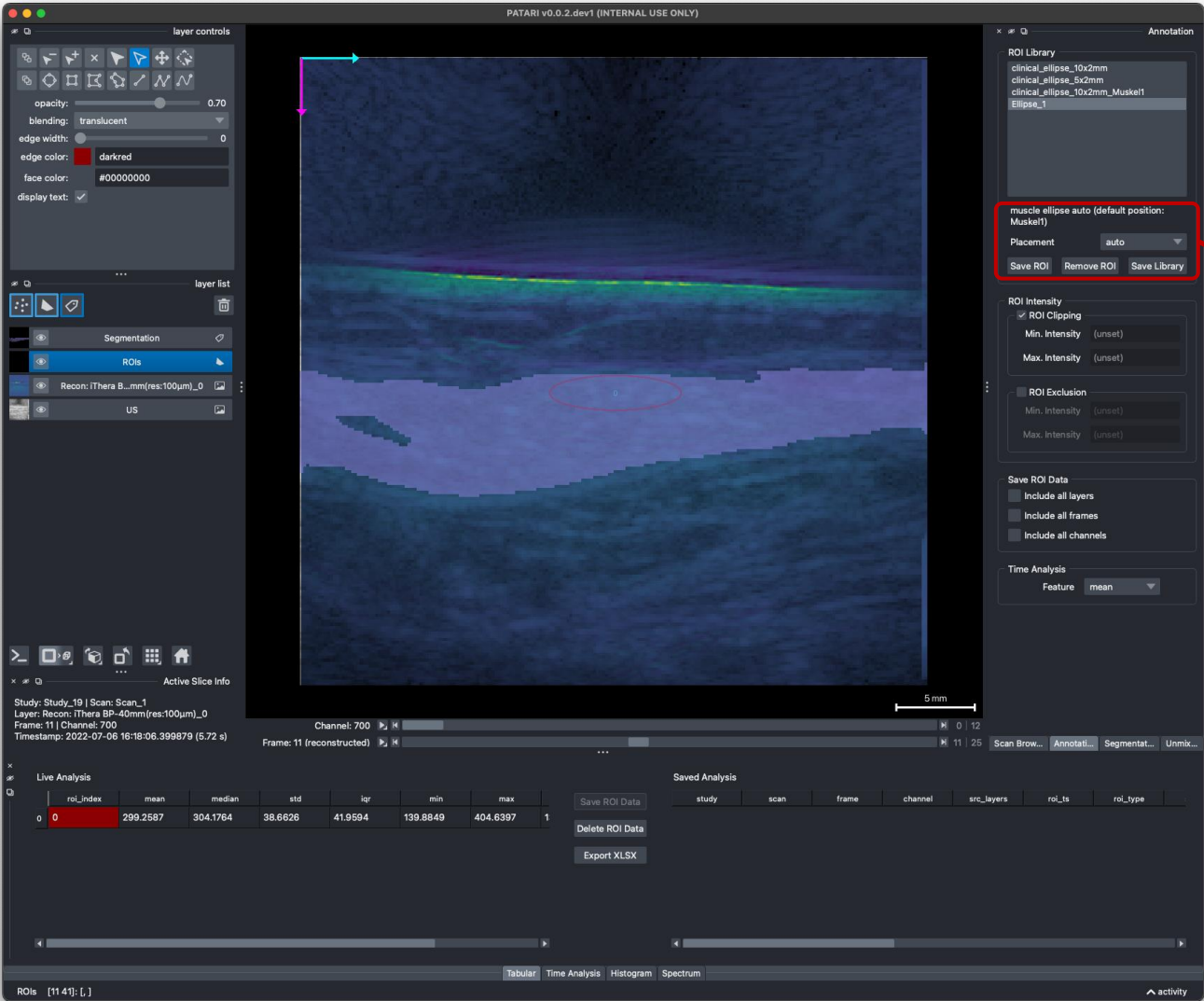


From the ROI Library



Saving ROI with Class ID

When saving a ROI to the ROI Library using the Save ROI button, you can provide a *Position*. You can set this position to a Class ID that is supported by the segmentation model



Auto ROI from Library

If a ROI in the Library has a position attribute that matches a class ID in your segmentation map, the *Placement* dropdown will switch to *auto*. If you double click such a ROI, it will automatically be placed in the corresponding tissue in your segmentation map.

If this behavior is not wanted, *Placement* can be set to static, to place to ROI at its fixed position from where it was initially saved.

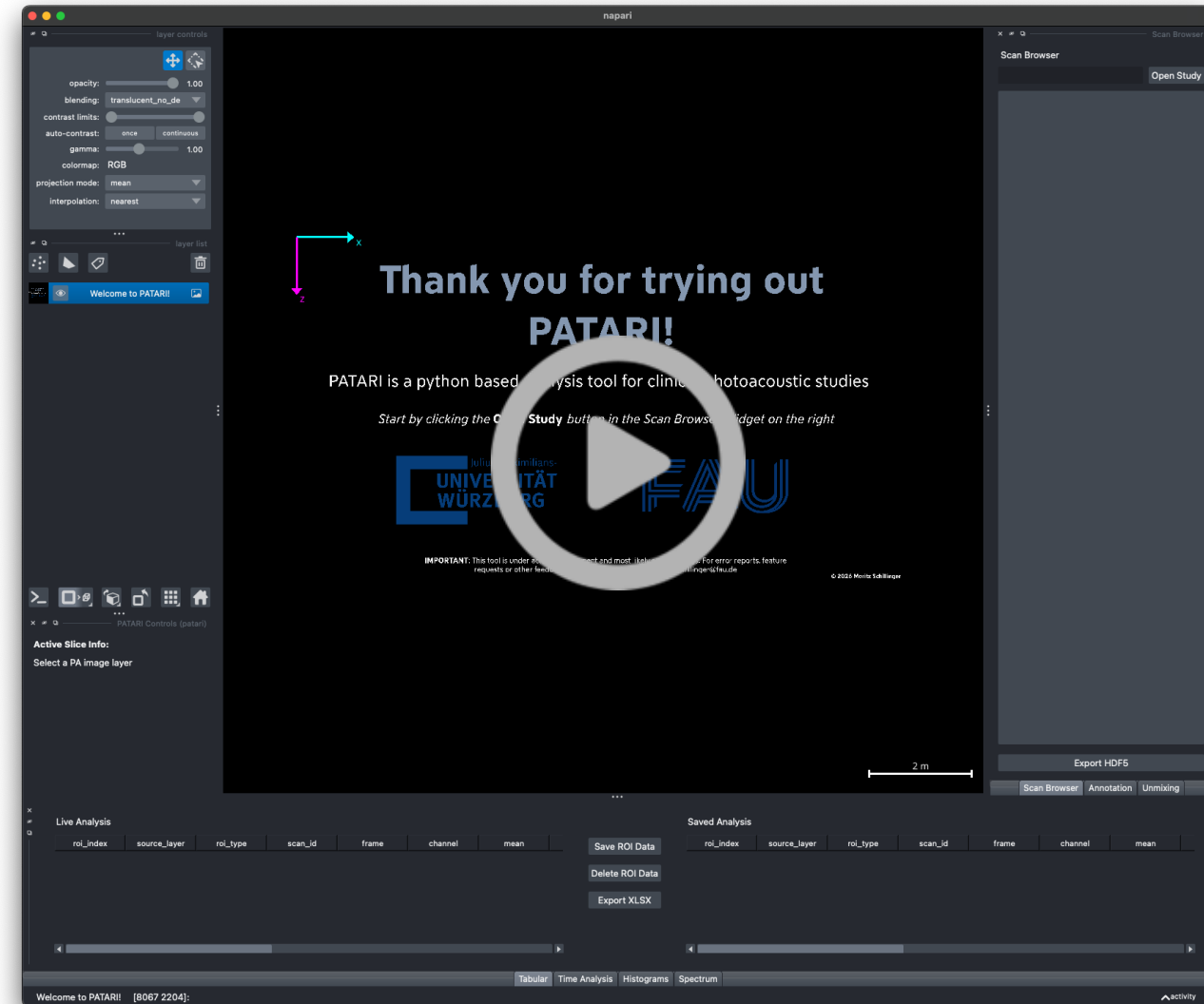
Example Workflows

Disclaimer: Some of the screenshots might be from an earlier version and therefore look slightly different.

Example Workflow

Manual ROI Placement and Using the ROI Library

1. Open Study folder (first scan is selected automatically)
2. Scroll to target channel and frame using the sliders
3. Select the ROI layer and draw the desired ROI shape that should be used for the analysis
4. Save the ROI to the ROI library with a given name and description
5. Store the ROI data in the Saved Table by clicking Save ROI Data
6. Continue to the next scan, scroll to target channel and frame
7. Place the ROI by double clicking on its name in the ROI Library
8. Move the ROI to the correct location
9. Store the ROI data in the Saved Table by clicking Save ROI Data
10. Continue for all scans, open new study folder, repeat ...
11. Finally, export all the saved ROI data to xlsx by clicking on the Export XLSX button
12. Optional: permanently add the ROI to the ROI Library by clicking Save Library

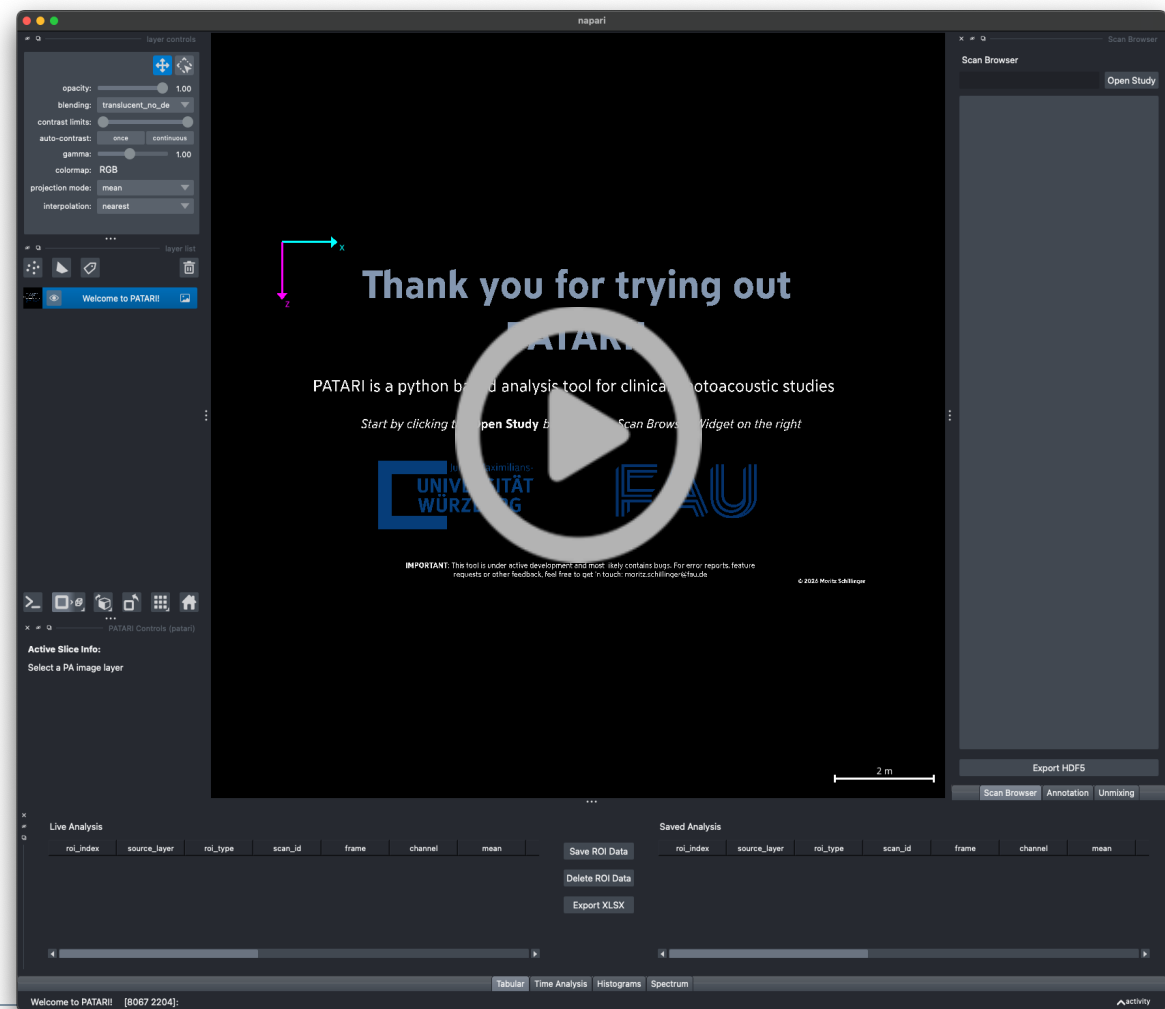


Example Workflow

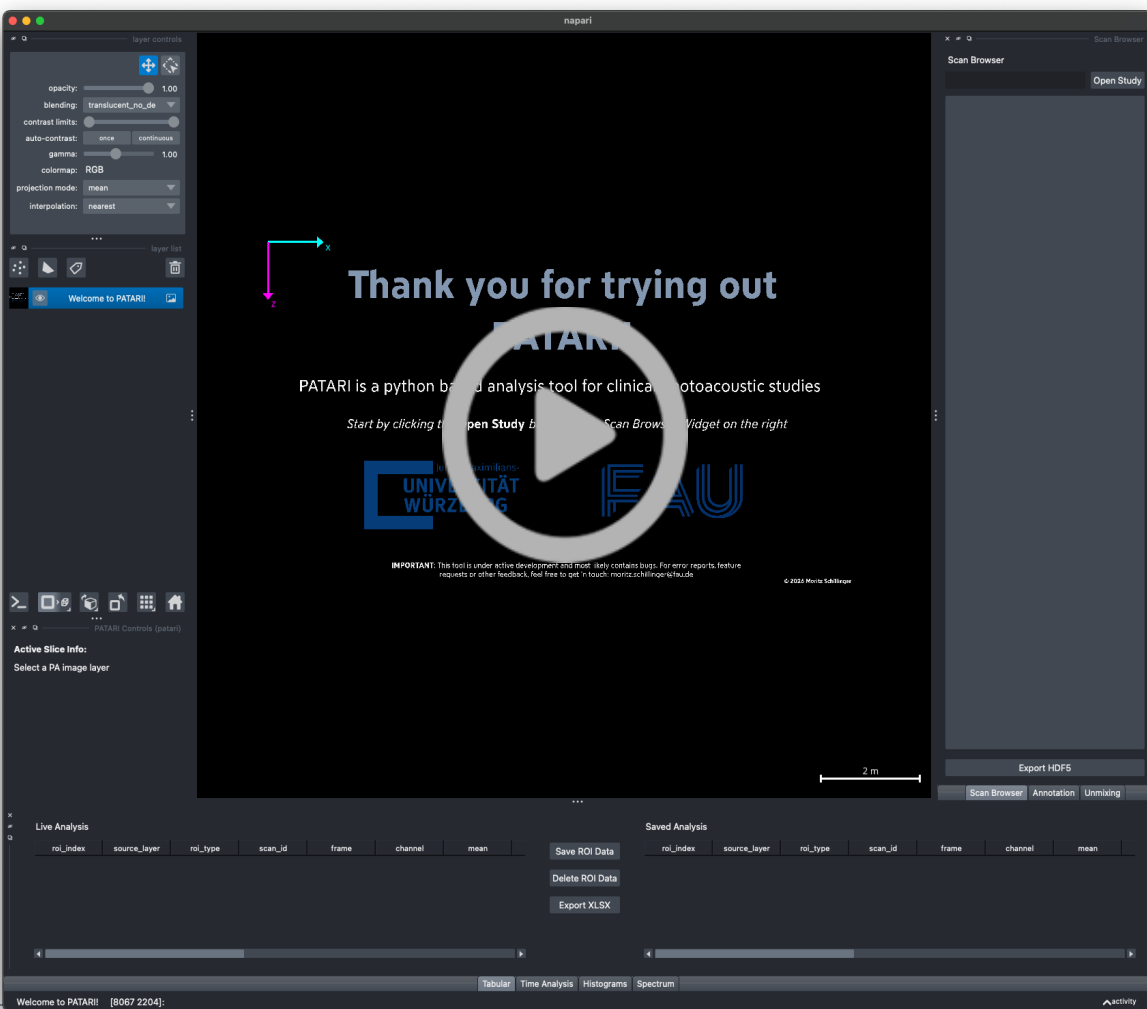


Tissue Segmentation and Unmixing

Tissue Segmentation



Unmixing



Example Workflow

Automatic ROI Placement

1. Open first scan
2. Configuration:
 - Configure segmentation settings
 - Configure unmixing settings
 - Draw the default ROI and save it to the library, using the desired tissue class ID
3. Perform automatic analysis, use keyboard shortcuts:
 - Segmentation: CMD – Shift – T
 - Unmixing: CMD – Shift – U
 - Save ROI Data: CMD – Shift - S
4. Move on to the next scan
5. If a ROI is placed wrongly, adjust it manually
6. Export the resulting table

